

# Find The Story Hidden in Your Data

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## Case study:

You are a junior data scientist in a company that just received a mysterious dataset  
No one knows what patterns exist, what factors influence outcomes, or what decisions can be made.

Your mission is to **investigate, understand, and build a predictive solution using SAS**

## Dataset Requirements

Students must choose their own dataset

### The dataset must:

- Contain at least 5000+ rows
- Include both categorical and numerical features
- Contain missing values
- Include a target variable for prediction
- Represent a real-world problem

Possible domains include: **customer behavior, healthcare, e-commerce, education, or transportation.**

**Team Members: 5 members**

**Deadline for early submission : 2/5/2026**

**Deadline for late submission : 7/5/2026**

**Submission Form: <https://forms.gle/y5MCjjbzk6rfQbaw8>**

## Your Mission

### 1. Understand the Data

Explain what each column represents, define the problem, and explain its importance.

### 2. Investigate the Data

Explore patterns, distributions, and relationships between variables.

### 3. Fix the Data

Identify data issues and justify how you handle them.

### 4. Create New Insights

Create new features and explain why they are useful.

### 5. Build a Model

Train and evaluate a predictive model using SAS.

### 6. Explain the Model

Identify important features and explain their impact.

### 7. Make It Useful

Explain how your solution helps real-world decision making.

## Final Output

Submit a **report**, **SAS code**, **visualizations**, and **prediction results**

## Constraints

- Do NOT follow tutorials blindly
- Do NOT copy solutions
- You must justify every decision
- Creativity is important

## Resources

### SAS Course Book:

<https://davidhason.com/wp-content/uploads/2024/02/SAS-Course-Book-All-in-One.pdf>

### SAS Programming Course:

<https://learn.sas.com/course/view.php?id=6345>

### Useful article:

<https://medium.com/@siddhartha.pachhai/kaggle-machine-learning-challenge-done-using-sas-a31605474b46>

## MINIMUM REQUIREMENTS

- Use **at least 1 classification model**
- Use **at least 4 features after engineering**
- Show **before/after cleaning**
- Include **at least 8 visualizations**
- Use **basic SAS procedures (PROC IMPORT, PROC MEANS, PROC REG / LOGISTIC)**

## Suggested team roles

- **Problem Owner** → defines the problem & goal
- **Data Explorer** → analyzes data & finds patterns
- **Data Cleaner** → fixes missing/dirty data
- **Feature Engineer** → creates useful new features
- **Model Builder** → trains model & makes prediction